Graham P. Taylor

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RESEARCH INTERESTS

Atmospheric Circulation, Climate Change Impacts, Extreme Weather, Downscaling/Bias-Correction

ACADEMIC EMPLOYMENT

Research Associate 2025–present
Cornell University

Postdoctoral Researcher 2024–2025
University Corporation for Atmospheric Research / NOAA Geophysical Fluid Dynamics Laboratory

Graduate Research Assistant 2019–2024
Department of Geography, Portland State University

Student Intern, Uncertainty Quantification and Statistical Analysis 2023
NASA Jet Propulsion Laboratory

EDUCATION

Ph.D., Earth, Environment, and Society 2024
Portland State University, Portland, Oregon

M.S., Geography 2021
Portland State University, Portland, Oregon

B.S., Physics 2013
St. Olaf College, Northfield, Minnesota

PEER-REVIEWED PUBLICATIONS

Taylor, G. P., K. Dixon, L. Sun, N. Zenes, S. Hartke, F. Lehner, A. Newman, E. Gutmann, R. McCrary . *How Observation-Based Data Influence Uncertainty in Local Climate Projections*. Journal of Applied Meteorology and Climatology, 65, 9, https://doi.org/10.1175/JAMC-D-25-0146.1

Loikith, P. C., S. Bigalke, **G. P. Taylor,** D. A. Kalashnikov, D. Singh, and A. Laverty (2026): *A climatology of heat domes over North America*. Weather and Climate Extremes, 53, 100913, https://doi.org/10.1016/j.wace.2026.100913.

Taylor, G. P., P. C. Loikith, H. Lee, S. Rahimi, and A. Hall (2025). *Historical and Future Autumn Rain and Wind Onset Over Western North America Using Regional Climate Models*. Journal of Geophysical Research: Atmospheres, 130. https://doi.org/10.1029/2025JD044267

Taylor, G. P., P. C. Loikith, H. Lee, B. Lintner, and C. M. Aragon (2023). *Projections of Large-Scale Atmospheric Circulation Patterns and Associated Temperature and Precipitation over the Pacific Northwest Using CMIP6 Models*. Journal of Climate, 36, 7257–7275. https://doi.org/10.1175/JCLI-D-23-0108.1

Taylor, G. P., P. C. Loikith, C. M. Aragon, H. Lee, and D. E. Waliser (2022). *CMIP6 Model Fidelity at Simulating Large-Scale Atmospheric Circulation Patterns and Associated Temperature and Precipitation over the Pacific Northwest*. Climate Dynamics, 60, 2199–2218. https://doi.org/10.1007/s00382-022-06410-1

Loikith, P. C., D. Singh, and **G. P. Taylor** (2022). *Projected Changes in Atmospheric Ridges over the Pacific–North American Region Using CMIP6 Models*. Journal of Climate, 35, 5151–5171. https://doi.org/10.1175/JCLI-D-21-0794.1

CONFERENCE PRESENTATIONS

Taylor, G. P., K. Dixon, L. Sun, N. Zenes, S. Hartke, F. Lehner, A. Newman, E. Gutmann, R. McCrary (2025). *Uncertainty Propagation from Observation-Based Data to Statistically Downscaled Climate Projections*. Fall Meeting of the American Geophysical Union, New Orleans, LA. (Poster)

Taylor, G. P., P. C. Loikith, and H. K. Lee (2024). *Projections of Change in the Timing of Fall Rain and Wind Over the Western United States*. Winter Meeting of the American Meteorological Society, New Orleans, LA. (Poster)

Taylor, G. P., P. C. Loikith, and H. K. Lee (2023). *Projections of Change in the Timing of Fall Rain and Wind Over the Western United States*. Fall Meeting of the American Geophysical Union, San Francisco, CA. (Talk)

Taylor, G. P., P. C. Loikith, and H. K. Lee (2023). *Future Conditions of Wind and Rain Associated with Fall Wildfire Conditions in CMIP6 Models over Western North America*. International Conference on Regional Climate (ICRC-CORDEX), Trieste, Italy. (Poster)

Taylor, G. P., P. C. Loikith, and C. Aragon (2022). *Projections of Large-Scale Atmospheric Circulation Patterns and Associated Temperature and Precipitation Anomalies over the Pacific Northwest Using CMIP6 Models*. Fall Meeting of the American Geophysical Union, Chicago, IL. (Poster)

Taylor, G. P., P. C. Loikith, and C. Aragon (2021). *Projections of Large-Scale Atmospheric Circulation Patterns and Associated Temperature and Precipitation Anomalies over the Pacific Northwest Using CMIP6 Models*. Fall Meeting of the American Geophysical Union, New Orleans, LA. (Talk)

Taylor, G. P., P. C. Loikith, and C. Aragon (2021). *Projections of Large-Scale Meteorological Patterns, Temperature, and Precipitation over the Pacific Northwest Using CMIP6 Models*. 11th Northwest Climate Conference. (Talk)

Taylor, G. P., P. C. Loikith, and C. Aragon (2020). *Projections of Future Large-Scale Meteorological Patterns, Temperature, and Precipitation over the Pacific Northwest*. Virtual Fall Meeting of the American Geophysical Union. (Poster)

Taylor, G. P., P. C. Loikith, C. M. Aragon, K. Heyn, K. Anderson, and B. Beal (2019). *Assessing Climate Change Impacts on Precipitation Over Bull Run Watershed*. Northwest Climate Conference, Portland, OR. (Poster)

INVITED TALKS

2025 UCAR, CPAESS Discovery Seminar – The Role of Observation-Based Data in Local Climate Projection Uncertainty.

2025 Rutgers, Department of Environmental Science Seminar Series.

2024 Center for Western Weather and Water Extremes, Scripps Institution of Oceanography, University of California San Diego. Science Friday Speaker Series.

2023 Portland State University, Geography Department Speaker Series.

OTHER PRESENTATIONS

2023 Jet Propulsion Laboratory, Uncertainty Quantification and Statistical Analysis Seminar Series.

2022 Portland State University, Student Research Symposium.

2022 Portland State University, Three Minute Thesis.

AWARDS

Travel Award, International Conference on Regional Climate (ICRC-CORDEX) and 11th Workshop on the Theory and Use of Regional Climate Models, Trieste, Italy, 2023.

PROFESSIONAL AFFILIATIONS

American Geophysical Union, 2020–present

American Meteorological Society, 2020–present

SERVICE

Journal Reviewer: *Climate Dynamics*, *International Journal of Climatology, Climatic Change, Scientific Reports*

SCIENCE COMMUNICATION AND OUTREACH

Guest Author, Early Career Climate Network. "What Do Climate Models Tell Us About the Future of Ridges over the Pacific Northwest?" 2022.

Interviewee, KOIN Podcast Special: Climate in Crisis. 2021.

TECHNICAL SKILLS

Python, R, Unix/Linux, high-performance computing, machine learning